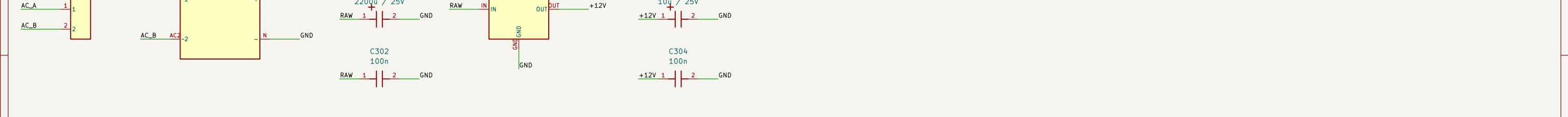


CTRL BOARD V0.6 / FIRST CIRCUIT DRAFT 2026-10-01 / UPC1237 protector + 12 V aux + soft-start timer / NO HARDWARE MEASUREMENT

Supply: T1 12 VAC winding (12.27 V no-load measured 2026-10-01, 1 A label) -> bridge -> 2200u -> 7812 -> +12V for relays, logic and UPC1237.

GND of this board ties to the PSU GND star at ONE point (J309). Relay coil current never flows through the amplifier signal ground (docs/13).

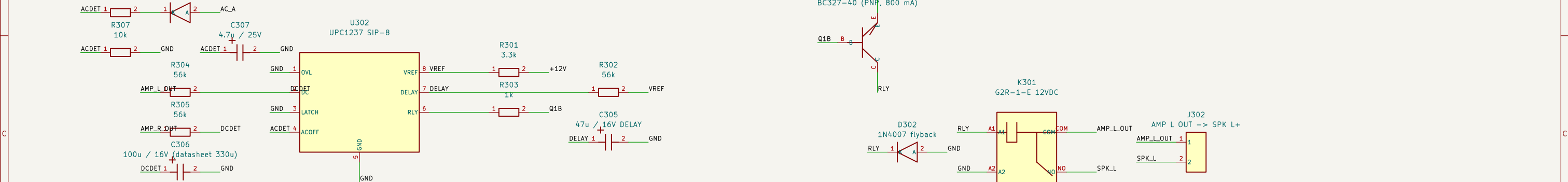
[illegible]

RAW -15-17 VDC at 150 mA (2 x 44 mA coils + bypass coil 44 mA + logic). 7812 dissipates ~1 W: bolt to chassis. Ripple 2200u @150 mA -0.6 Vpp (docs/14 6.5).

B. UPC1237 PROTECTOR (UTC QW-R107-065.B)
Thresholds: pin 2 +0.62 V / -0.17 V (DC), pin 4 0.74 V (AC-off), pin 8 = 3.4 V reference, pin 6 sinks <= 80 mA.

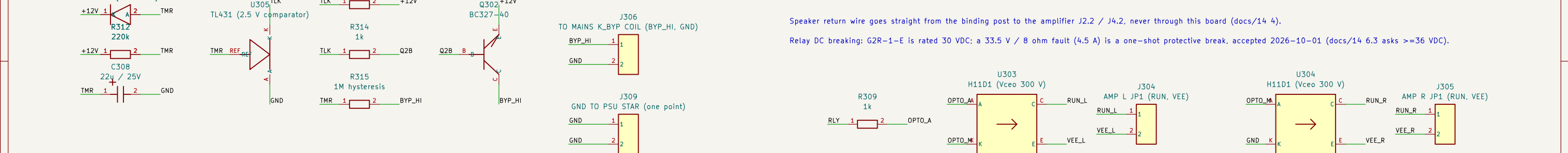
R306 D301
 56k 1N4148

Q18 1 2 +12V
 Q301 +12V



DC trip (one channel, other at 0 V): +1.24 V / -0.34 V at the speaker output. 56k|56k x 100u = 2.8 s: 30 V fault trips in ~20 ms; 25 Vpk at 10 Hz leaves 0.07 V at pin 2 (<0.17 V).
AC-off: 17 Vpk x 10k/66k = 2.6 V at pin 4 while mains is on (droop between peaks to 1.8 V); after loss 10k x 4.7u decays below 0.74 V in ~60 ms (docs/14 6.4 budget <60 ms).

D. SOFT-START BYPASS TIMER: K_BY (on MAINS board) pulls in ~1.1 s after +12V (docs/14 6.2 recommends 1 s)



t = $220k \times 22\mu \times \ln(12/9.5) = 1.1$ s. TL431 snaps at 2.5 V (sharp, no relay chatter); R315 adds -0.3 V hysteresis; D305 empties C308 within ms at power-off so a quick off-on restarts the NTC limit.

E. FRONT PANEL LEDS: green = speakers ON (from RLY), red = +12V present but relay OFF (returns into RLY)

R310 1k GREEN
 RLY 1 2 LED_G
 5V 1k 1mA

R311 1.5k RED
 +12V 1 2 LED_R
 5V 1.5k 1mA

R312 1k GREEN
 5V 1k 1mA

R313 1.5k RED
 5V 1.5k 1mA

R314 1k GREEN
 5V 1k 1mA

R315 1.5k RED
 5V 1.5k 1mA

R316 1k GREEN
 5V 1k 1mA

R317 1.5k RED
 5V 1.5k 1mA

R318 1k GREEN
 5V 1k 1mA

R319 1.5k RED
 5V 1.5k 1mA

R320 1k GREEN
 5V 1k 1mA

R321 1.5k RED
 5V 1.5k 1mA

R322 1k GREEN
 5V 1k 1mA

R323 1.5k RED
 5V 1.5k 1mA

R324 1k GREEN
 5V 1k 1mA

R325 1.5k RED
 5V 1.5k 1mA

R326 1k GREEN
 5V 1k 1mA

R327 1.5k RED
 5V 1.5k 1mA

R328 1k GREEN
 5V 1k 1mA

R329 1.5k RED
 5V 1.5k 1mA

R330 1k GREEN
 5V 1k 1mA

R331 1.5k RED
 5V 1.5k 1mA

R332 1k GREEN
 5V 1k 1mA

R333 1.5k RED
 5V 1.5k 1mA

R334 1k GREEN
 5V 1k 1mA

R335 1.5k RED
 5V 1.5k 1mA

R336 1k GREEN
 5V 1k 1mA

R337 1.5k RED
 5V 1.5k 1mA

R338 1k GREEN
 5V 1k 1mA

R339 1.5k RED
 5V 1.5k 1mA

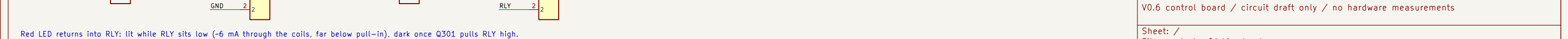
R340 1k GREEN
 5V 1k 1mA

R341 1.5k RED
 5V 1.5k 1mA

R342 1k GREEN
 5V 1k 1mA

R343 1.5k RED
 5V 1.5k 1mA

R344 1k GREEN
 5V 1k 1mA



NOT DRAWN YET: single-rail-loss detector (docs/13 P5, optional), over-temperature (P6, optional). Component values are first-pass; every timing and threshold is to be measured per docs/14 9 before the board goes into the chassis.

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